TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	65.9 ± 2.70
Correlation Stability	90.4 ± 0.49
Distribution Consistency	68.9 ± 5.52
Entropy Preservation	49.6 ± 2.11
Mean Z Score Drift	0.3 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	25.8 ± 3.92
Data Density	100.0 ± 0.00
Coverage Score	62.8 ± 3.60
Feature Accuracy	21.0 ± 1.48
Novelty Score	0.6 ± 0.03

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Dataset Audit: thoracic_surgery.csv

1. Dataset Portrait

Attribute	Value
Rows	470
Columns	17
Numeric features	17
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
DGN	3.096	2.784	-0.312	10.1%	Warn
PRE4	3.282	3.184	-0.098	3.0%	OK
PRE5	4.569	3.275	-1.294	28.3%	Drift
PRE6	0.781	0.830	+0.049	6.2%	OK
PRE7	0.066	0.000	-0.066	100.0%	Drift
PRE8	0.145	0.030	-0.114	79.1%	Drift
PRE9	0.066	0.004	-0.062	94.3%	Drift
PRE10	0.687	0.795	+0.108	15.7%	Warn
PRE11	0.166	0.064	-0.102	61.2%	Drift
PRE14	11.736	11.758	+0.021	0.2%	OK
PRE17	0.074	0.019	-0.056	74.6%	Drift
PRE19	0.004	0.000	-0.004	100.0%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	65.9 ± 2.70
Correlation Stability (*2)	90.4 ± 0.49
Distribution Consistency (*3)	68.9 ± 5.52
Entropy Preservation (*4)	49.6 ± 2.11
Mean Z Score Drift (*5)	0.3 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	25.8 ± 3.92

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	62.8 ± 3.60
Feature Accuracy (*10)	21.0 ± 1.48
Novelty Score (*11)	0.6 ± 0.03

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	N/A	0.4670212765957447	-0.3882978723404255	*
F1	N/A	N/A	0.16560308434225524	0.16560308434225524	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.851063829787234	N/A	0.553191489361702	-0.2978723404255319	*
F1	0.037675637056442016	N/A	0.20122575052980612	0.1635501134733641	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8521276595744682	N/A	0.6659574468085105	-0.18617021276595744	*
F1	N/A	N/A	0.20270605982039736	0.20270605982039736	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	N/A	0.5446808510638298	-0.31063829787234043	*
F1	N/A	N/A	0.24902618512954855	0.24902618512954855	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8425531914893618	N/A	0.4755319148936169	-0.3670212765957447	*
F1	0.0125	N/A	0.16271059573618002	0.15021059573618004	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.8127659574468085	N/A	0.5957446808510639	-0.21702127659574466	*
F1	0.11598151904439935	N/A	0.19746075315842757	0.08147923411402822	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8382978723404257	N/A	0.6648936170212767	-0.1734042553191489	*
F1	0.07546915644741732	N/A	0.18899362016283505	0.11352446371541776	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.21063829787234045	N/A	0.22234042553191488	0.011702127659574468	
F1	0.256855620661189	N/A	0.2070899075091479	-0.04976571315204107	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.847872340425532	N/A	0.5106382978723404	-0.3372340425531914	*
F1	N/A	N/A	0.17987947733158408	0.17987947733158408	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8138297872340425	N/A	0.6968085106382979	-0.11702127659574468	*
F1	0.11972345964178996	N/A	0.20456302862328685	0.08483956898149692	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.88	2.16	1.0	0.0	0.0	0.0	1.0	1.0	14.0
2	3.0	3.4	1.88	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.76	2.08	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.68	3.04	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.44	0.96	2.0	0.0	1.0	0.0	1.0	1.0	11.0
6	3.0	2.48	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	4.36	3.28	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	2.0	3.19	2.5	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	3.16	2.64	2.0	0.0	0.0	0.0	1.0	1.0	11.0
10	3.0	2.32	2.16	1.0	0.0	0.0	0.0	1.0	0.0	11.0
11	3.0	2.56	2.32	0.0	0.0	1.0	0.0	1.0	0.0	12.0
12	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
13	3.0	3.0	2.36	1.0	0.0	0.0	0.0	1.0	1.0	11.0
14	2.0	3.98	3.06	2.0	0.0	0.0	0.0	1.0	1.0	14.0
15	3.0	1.96	1.4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
16	3.0	4.68	4.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
17	2.0	2.21	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
18	2.0	2.96	1.67	0.0	0.0	0.0	0.0	0.0	0.0	12.0
19	3.0	2.6	1.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
20	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	3.4960366239399114	2.3610935242226017	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.5409441823287584	2.046142120928468	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	2.0	2.801237317514409	1.6086612226008634	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.722087408028447	1.8668986008920496	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	2.0	3.1869757082329766	2.5129300944310637	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	2.0	2.257241093744828	1.4023526164156865	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	2.68	2.16	0.0	0.0	0.0	0.0	0.0	0.0	12.0
8	4.0	2.32	1.76	1.0	0.0	1.0	0.0	1.0	1.0	11.0
9	2.0	2.72	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	3.096430184054982	2.4045811283579264	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	2.0	3.060143563597483	2.3154573860520227	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	2.0	3.069964908906172	3.438189491906648	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	2.0	3.073401699071794	2.33053526561106	0.0	0.0	0.0	0.0	0.0	0.0	12.0
5	4.0	2.940912186580396	2.141329458664126	1.0	0.0	0.0	0.0	0.0	0.0	13.0
6	3.0	2.770651501678711	2.202379579654194	1.0	0.0	0.0	0.0	0.0	0.0	13.0
7	3.0	2.84	2.24	1.0	1.0	1.0	0.0	0.0	0.0	12.0
8	3.0	3.36	2.8	1.0	0.0	0.0	0.0	1.0	1.0	12.0
9	3.0	2.44	1.96	1.0	0.0	1.0	1.0	1.0	0.0	13.0
10	3.0	5.16	4.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	3.0747630351289716	1.7426935144411233	0.0	0.0	0.0	0.0	1.0	0.0	12.0
2	2.0	3.050888023044531	1.8461519001403641	0.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	4.251304446261575	3.1475272938812946	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	3.172096915036831	1.8325628961561269	1.0	0.0	0.0	0.0	1.0	1.0	12.0
5	3.0	3.375078029261169	2.546976873369976	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	4.0	3.536966012600723	2.578317287940604	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	4.0	4.24	3.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	3.87	2.68	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	5.0	3.67	76.8	0.0	1.0	1.0	0.0	1.0	0.0	12.0
10	3.0	3.8	2.84	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	3.477510276613322	2.722936726065896	1.0	0.0	0.0	0.0	0.0	0.0	12.0
2	3.0	3.588576248394248	2.903401279768384	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	3.793997866221228	2.952129391744744	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	3.472864809864686	2.216342807133687	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.133229533411974	2.313076450154303	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	3.328990053659692	2.025225593667351	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	1.84	1.28	1.0	0.0	0.0	0.0	1.0	1.0	11.0
8	5.0	4.95	4.12	1.0	0.0	0.0	0.0	0.0	1.0	11.0
9	3.0	2.92	2.28	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.7118376196255682	1.9204059509360794	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.7105833311377663	1.923541672155584	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	2.4816267378213093	1.8132957532435852	0.0	0.0	0.0	0.0	0.0	0.0	12.0
4	3.0	2.7295660459513504	1.876084885121624	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	2.5273158193486664	1.8562972417399213	0.0	0.0	0.0	0.0	0.0	0.0	12.0
6	2.0	2.8284461728681225	62.16252568626336	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	3.0	2.44	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	2.4	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	4.36	3.92	1.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	3.4	2.8	1.0	1.0	0.0	0.0	1.0	1.0	11.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	4.0	3.783905459311364	2.9535589688686694	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	4.0	3.947376146344686	3.122463608133196	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	3.604931995578775	2.49232998894693	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	5.0	4.661088842541073	3.85593520189523	1.0	0.0	0.0	0.0	1.0	0.0	11.0
5	3.0	4.328757113076312	3.491131875973768	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	3.731936222062228	2.963185268565183	0.0	0.0	0.0	0.0	0.0	0.0	11.0
7	2.0	3.24	2.52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.4	1.64	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	3.24	52.3	0.0	0.0	0.0	0.0	0.0	0.0	12.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.5053819385660314	1.8428202153962259	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.514625846341575	2.060763418699125	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	2.8461055826950403	1.8854628897797732	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.742128895095777	1.8691254327884197	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	2.986074141428478	2.1975129599133463	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	2.780332432493142	2.0354134316612638	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	2.84	2.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	2.2	1.44	1.0	0.0	0.0	0.0	1.0	0.0	12.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.532418496327131	2.0671468174175676	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	3.0	2.519238366638681	2.0100329221009536	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	2.0	2.840720771650355	2.140600643041963	0.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.574722712948063	1.8505247458831184	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.206305184650877	2.0716640254396643	0.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	2.755013928555918	1.7971133270586896	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	2.76	2.28	0.0	0.0	0.0	0.0	0.0	0.0	11.0
8	4.0	5.03	79.3	1.0	0.0	0.0	1.0	0.0	0.0	11.0
9	2.0	2.16	1.56	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	11.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.455854575267245	1.1106184650388253	2.0	0.0	0.0	0.0	1.0	1.0	11.0
2	3.0	2.833748852189339	2.033748852189339	2.0	0.0	0.0	0.0	1.0	1.0	11.0
3	3.0	2.452236361347771	1.0762454328038225	2.0	0.0	0.0	0.0	1.0	1.0	11.0
4	3.0	2.722566515353613	1.922566515353613	2.0	0.0	0.0	0.0	1.0	1.0	11.0
5	3.0	3.303380890135837	2.964844816042895	0.0	0.0	0.0	0.0	0.0	0.0	11.0
6	3.0	2.189618783566019	1.829232915936433	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	3.44	3.13	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	3.64	3.12	1.0	0.0	0.0	0.0	1.0	0.0	12.0
9	3.0	2.56	60.9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	2.8	3.36	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	4.37812945270225	3.4069061689157483	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	3.0	3.250470040184394	2.599665508500134	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	4.371970474996691	3.3637933249768324	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.619186899344642	2.672153353308424	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.545400361793563	2.542981279718015	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	4.139086306997129	3.144601284933723	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	3.48	2.52	1.0	0.0	0.0	0.0	1.0	0.0	14.0
8	3.0	4.48	3.48	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	3.0	2.63	67.3	1.0	0.0	0.0	1.0	1.0	0.0	11.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0